

Name: _____

Per _____

Chapter 7 Practice Test

7.1 & 7.2: I can transform and create exponential functions to model situations.

1. Graph the following functions on the graph.

Show at least 3 points per graph and label each.

$$f(x) = 3^x + 0$$

Asymptote $y = 0$

$$g(x) = \frac{1}{2} \cdot 3^{x-2}$$

 $\leftarrow \text{exp} = 0$ $\text{exp} = 1$

$$x - 2 = 0$$

$$x - 2 = 1$$

$$x = 2$$

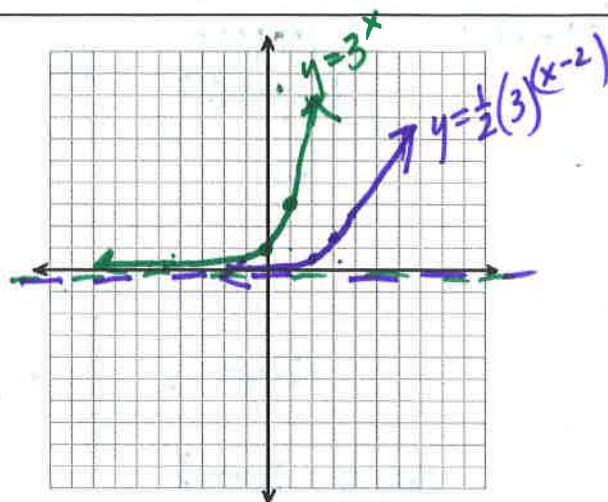
$$x = 3$$

$$y = \frac{1}{2}(3)^{2-2}$$

$$y = \frac{1}{2}(3)^{3-2}$$

$$y = \frac{1}{2}$$

$$y = \frac{3}{2}$$



2. The current student enrollment of Paly is 2000 students and expected to decline 2% each year for the next ten years. If these projections are correct, what will Paly's enrollment be in ten years? Create mathematical model and round to the nearest student.

$$A = P(1-r)^t$$

$$2\% = .02$$

$$A = 2000(1 - .02)^{10}$$

$$A = 1634 \text{ students}$$

3. A typical credit card charges an annual rate of 25.24% compounded monthly on all purchases. Let's say you bought a 2019 Tesla Model 3 (\$44,000.00) and fail to send a single payment for five years. Barring any penalties that may be incurred, how much would you owe after five years?

$$A = P\left(1 + \frac{r}{n}\right)^{nt} = 44,000\left(1 + \frac{.2524}{12}\right)^{(12 \times 5)}$$

$$= \$153,407.97$$

4. You put \$10,000 in a money market account that pays 2.3% compounded quarterly. How much do you have after 12 years?

$$A = P\left(1 + \frac{r}{n}\right)^{nt} = 10,000\left(1 + \frac{.023}{4}\right)^{(4 \times 12)}$$

$$= \$13,168.07$$

7.3: I can simplify logarithmic expressions and graph logarithmic functions.

$$\log_2 16 \text{ or } 2^4 = 16$$

Simplify the following:

5. $\log 1000000$ $\log_2 16 = 4$	6. $\log 1000000$ $10^6 = 1000000$	7. $\log_9 81$ 2	8. $\log_5 \frac{1}{25}$ -2	9. $\log 0$ undefined	10. $\log 1$ 0
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11. Graph the following functions on the graph.

Show at least 3 points per graph and label each.

$$f(x) = \log_2 x$$

$$g(x) = \log_2(x+3) - 1$$

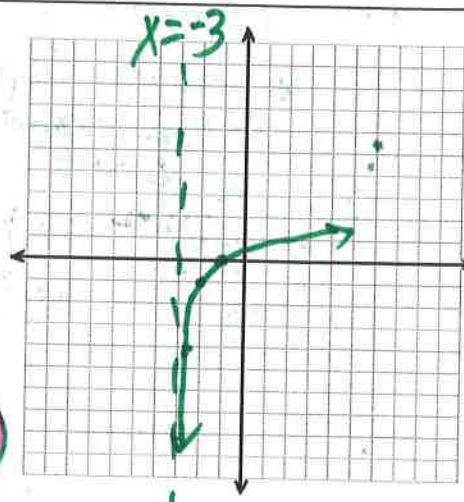
Left +3 Down 1

shape

asym: $x = -3$

① $() = \text{base}$
 $(x+3) = 2$
 $x = -1$
 $y = \log_2(-1+3) - 1 = 1 - 1 = 0$
 $(-1, 0)$

② $() = 1$
 $x+3 = 1$
 $x = -2$
 $y = \log_2(-2+3) - 1 = 0 - 1 = -1$
 $(-2, -1)$



7.4: I can use the properties of logarithms to simplify and expand expressions.

12. Expand $\log x^2 y^3$

$$\log x^2 + \log y^3$$

$$2 \log x + 3 \log y$$

13. Expand $\log \frac{16(m-2)^3}{np}$

$$\log 16(m-2)^3 - (\log(n \cdot p))$$

$$\log 16 + \log(m-2)^3 - (\log n + \log p)$$

$$\log 16 + 3 \log(m-2) - \log n - \log p$$

14. Condense $\log(x) - \log(5y) + \log(2z)$

$$\log\left(\frac{x}{5y}\right) + \log(2z)$$

$$\log\left(\frac{x}{5y}\right) + \log\left(\frac{2z}{1}\right)$$

$$\log\left(\frac{2xz}{5y}\right)$$

remember

15. Condense $\log(25) + \log(2) - \log(3)$

$$\log 50 - \log 3$$

$$\log\left(\frac{50}{3}\right)$$

7.5: I can solve exponential and logarithmic equations.

For each of the following equations, find all possible values for x that solve it. If necessary, round to the nearest thousandth.

16. $8^{3x} = 32$

$$\log 8^{3x} = \log 32$$

$$3x \log 8 = \log 32$$

$$x = \frac{\log 32}{3 \log 8} \approx \boxed{.556}$$

thousandths
tenths hundredths

17. $4^x = 13$

$$\log 4^x = \log 13$$

$$x \log 4 = \log 13$$

$$x = \frac{\log(13)}{\log(4)} \approx \boxed{1.850}$$

18. $15^{2x-1} = 1000$

Rewrite from exponential

$$\log 15^{2x-1} = \log 1000$$

$10^3 = 1000$

$$(2x-1) \log 15 = 3$$

$\log$ is very important!

$$2x \log 15 - 1 \log 15 = 3$$

$$2x \log 15 = 3 + \log 15$$

$$x = \frac{(3 + \log 15)}{(2 \log 15)} \approx \boxed{1.775}$$

19. $\log_3(x-4) = 2$

$$3^2 = (x-4)$$

$$9 = x-4$$

$$x = 13$$

check is $(x-4) > 0$
 $13-4 > 0$
yes!

20. $2 \log(x-2) = \frac{2}{2}$ Divide by 2

$$\log_{10}(x-2) = 1$$

$$10^1 = x-2$$

$$x = 12$$

check!
 $x-2 > 0$
 $12-2 > 0$
yes!

21. $\log(x+9) + \log(x) = 1$

Condense

$$\log(x+9)(x) = 1$$

$$\log(x^2+9x) = 1$$

$$10^1 = x^2+9x$$

$$0 = x^2+9x-10$$

$$0 = (x+10)(x-1)$$

$$x = -10 \quad x = 1$$

check $x = -10$
 $x+9 > 0$
 $-10+9 > 0$
NO
And
 $x > 0$
 $-10 > 0$ NO

check $x = 1$
 $x+9 > 0$
 $10 > 0$
yes
 $x > 0$
 $1 > 0$
yes

22. The Palo Alto squirrel population increases 3.5% every year. If there are 5,000 squirrels in Palo Alto today, how long will it take to reach 8,000? $\hookrightarrow .035$

$$A = P(1+r)^t$$

$$8000 = 5000(1+.035)^t \rightarrow \text{Divide by 5000}$$

$$1.6 = (1.035)^t$$

$$\log 1.6 = \log(1.035)^t$$

$$\log(1.6) = t \log(1.035)$$

$$t = \frac{\log(1.6)}{\log(1.035)} = t$$

$$t = 13.662 \text{ years}$$

23. You put \$40,000 on a credit card with 19% interest compounded continuously. If you don't pay off any of your debt, how long will it take to reach \$100,000? (This credit card has no other fees or penalties)

P $\rightarrow$ $A = P(1 + \frac{r}{n})^{nt}$

$$100,000 = 40,000(1 + \frac{.19}{2})^{2t}$$

semi annually
 $\rightarrow$ twice a year
 $19\% = .19$

$$2.5 = (1.095)^{2t}$$

$$\log(2.5) = \log(1.095)^{2t}$$

$$\log(2.5) = 2t \log(1.095)$$

$$\frac{\log(2.5)}{(2 \log(1.095))} = t$$

$\uparrow$
 Don't forget

$$t = 5.04819$$

$$t = 5.048 \text{ years}$$